

Jonathan Lozano

PhD Candidate in Theoretical Physics

Weinberg Institute for Theoretical Physics
The University of Texas at Austin
✉ jonathanloz@utexas.edu
📄 jonathanloz.github.io
ORCID: 0000-0002-9638-5173

Education

- 2023–present **Ph.D. in Physics**, *The University of Texas at Austin, USA*
Advisor: Dr. Antonios Alvertis.
GPA: 3.9/4.00. Research in condensed matter and many-particle physics.
- 2016–2021 **B.Sc. in Physics**, *National Autonomous University of Mexico (UNAM), Mexico City*,
GPA: 9.39/10.0 — *High Honors*
Advisor: Dr. Manuel Torres Labansat. Coursework included two graduate-level courses in quantum field theory and a graduate course in differential geometry and topology for physicists.
Thesis: Spontaneous symmetry breaking and extended field configurations in a scalar theory subject to a potential with two families of vacuum states.

Research Interests

Condensed matter physics · Many-particle physics · Quantum phenomena ·
Tensor networks for strongly correlated systems · Topological and non-topological
solitons · Non-perturbative phenomena

Publications

Refereed journal articles

- 2026 **J. Lozano-Mayo**. Generalized virial identities: radial constraints for solitons, instantons, and bounces. *J. High Energ. Phys.* **04** (2026) 058. doi:10.1007/JHEP04(2026)058
- 2026 **J. Lozano-Mayo** and M. Torres-Labansat. Multi-kinks and composite oscillons in a commensurable and non degenerate double sine-Gordon model. *J. High Energ. Phys.* **09** (2026) 218. doi:10.1007/JHEP09(2026)218
- 2021 **J. Lozano-Mayo** and M. Torres-Labansat. Kink solutions in a generalized scalar ϕ_G^4 field model. *J. Phys. Commun.* **5** (2021) 025004. doi:10.1088/2399-6528/abdd83

Submitted

- 2025 J. J. Ziegler, K. Freese, **J. Lozano**, G. Montefalcone. Explaining the “too massive” high-redshift galaxies in JWST data: a numerical study of three effects and a simple relation. *In revision*.

Collaboration authorship

- 2024–2025 *ALICE Collaboration*. Papers carrying my name under ALICE’s collective authorship policy are not listed individually.

Research Experience

- 2026–present **Exciton-polaron physics in the Holstein model**, *Weinberg Institute, UT Austin*, Alvertis group
Computing exciton binding energies in strongly coupled electron-phonon systems with tensor-network methods, in the regime where perturbative treatments of the coupling break down.

- 2025–2026 **Generalized virial identities for solitons, instantons and bounces**, *Weinberg Institute, UT Austin*, Independent
Derived a one-parameter family of virial identities for $O(n)$ -symmetric configurations that resolves Derrick’s global constraint into core and tail contributions. Verified them analytically for the Fubini-Lipatov instanton, the BPS monopole and the BPST instanton, and established their residuals as an accuracy diagnostic for numerical solutions. Sole-authored, published in JHEP.
- 2025 **Charm production across color environments**, *ALICE Collaboration, UT Austin*, Markert group
Measured h – D^0 azimuthal correlations in ALICE Run-3 Pb–Pb data to probe charm production in different color environments, and contributed ITS2 performance studies as collaboration service work.
- 2024 **High-redshift galaxy masses and top-heavy IMFs**, *Weinberg Institute, UT Austin*, Freese group
Modeled high-redshift galaxy spectral energy distributions with PÉGASE to determine where JWST observations remain consistent with Λ CDM.
- 2020–2022 **Solitons in scalar field theories with non-degenerate vacua**, *Institute of Physics, UNAM*
Established the existence of static n -kink structures in deformed potentials by asymptotic analysis, characterized the inter-kink binding forces numerically, and derived an analytical relation for the multi-kink energy. Renormalized the one-loop quantum mass correction of kinks in a generalized ϕ^4 potential. Basis of two refereed papers.

Awards & Distinctions

- 2024 **Biedenharn Endowment for Excellence** — Department of Physics, UT Austin.
- 2022 **DPG/IAPS Travel Grant** — awarded by the German Physical Society and the International Association of Physics Students to attend the PLANCKS international final at LMU Munich.
- 2022 **Winner, Mexican Tournament of Physics (TMF)** — placed 1st of 30 teams in the *Torneo Mexicano de Física*, a theoretical physics competition for bachelor’s and master’s students that serves as the Mexican PLANCKS preliminary.
- 2021 **Juan Manuel Lozano Mejía Diploma**, UNAM Institute of Physics — institute-wide distinction for the most outstanding undergraduate thesis research in physics.
- 2021 **Excellence Cluster Fellowship** — PRISMA, Johannes Gutenberg University Mainz.
- 2021 **PLANCKS 2021 (Porto): 7th of 50 teams** — international final of the theoretical physics competition, in Mexico’s first appearance.
- 2021 **Winner, Mexican Tournament of Physics (TMF)** — placed 1st nationally in the PLANCKS preliminary, earning a place on the first Mexican team to attend PLANCKS.
- 2021 **Honorific Mention**, B.Sc. thesis defense (UNAM) — highest distinction a thesis committee can award.
- 2019 **Research Assistantship**, UNAM Institute of Physics.
- 2017–2018 **Summer research and school scholarships** — Mexican Academy of Sciences (XXVIII Scientific Research Summer, 2018), ICF-UNAM (VII Experimental Physics School, 2018) and IF-UNAM (XXV Physics School, 2017).

Conference Presentations

- 2025 **J. L. Mayo**. h – D^0 angular correlations and Δ ROF performance studies at ITS2. ALICE-USA Meeting.

- 2021 **J. L. Mayo.** *Multi-solitones en teorías escalares de campo con vacíos no-degenerados: el modelo doble de sine-Gordon.* LXVI National Physics Congress, Mexican Physical Society.

Teaching & Academic Service

- 2026 **Teaching Assistant — Graduate Quantum Mechanics, UT Austin**
Mentored students and graded assignments for a graduate course on Quantum Mechanics.
- 2025 **Lecturer — Quantum Field Theory I, ICTP (International Centre for Theoretical Physics)**
Delivered an introductory quantum field theory course as part of the International Centre for Theoretical Physics outreach initiative.
- 2024 **Teaching Assistant — Advanced Particle Physics, ICTP**
Led problem-set discussions and graded assignments for a graduate course on advanced topics in quantum field theory.
- 2025 **Teaching Assistant — Waves and Optics, UT Austin**
Held weekly discussion sections and graded coursework spanning wave phenomena, optics, and electromagnetism.
- 2024–2025 **Teaching Assistant — Modern Physics, UT Austin**
Led weekly discussion sections and graded coursework spanning statistical mechanics, quantum mechanics, and special relativity.
- 2023 **Teaching Assistant — Electromagnetism Laboratory (PHY 105N), UT Austin**
Led two weekly laboratory sections and supported the experimental setups.
- 2022 **Teaching Assistant — Thermodynamics, UNAM**
Held problem-solving sessions and created and graded homework and exams.
- 2020 **Teaching Assistant — Nuclear and Subnuclear Physics, UNAM**
Mentored students and graded homework and exams for a course taught by Prof. Manuel Torres Labansat.
- 2022 **Referee, J. Phys. G: Nuclear and Particle Physics**
Reviewed submitted manuscripts.
- 2017 **Volunteer Instructor, UNAM School of Humanities and Sciences**
Taught physics and mathematics in an undergraduate preparation course.

Technical Skills

Programming	Julia, Python, C++, Mathematica, $\text{\LaTeX}$, HTML
Numerical methods	Tensor networks (DMRG), exact diagonalization, finite-difference PDE solvers
Scientific tools	ROOT/O ² (ALICE framework), QtiPlot, Origin, Tracker, ImageJ

Additional Training

- 2021–2025 Summer School on CMB Data Analysis (UT Austin, 2025); Modeling and Tools for Data Analysis in Science (UANL, 2022); Bad Honnef Physics School on Methods of Effective Field Theory and Lattice Field Theory (DPG, 2021); VII Mexican School on String Theory and Supersymmetry and Mexican Astro-Particle School (University of Guanajuato, 2021).

Languages

Spanish (native) · English (C1) · German (A2) · French (A2)